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TECHNOLOGIES



RESEARCH METHODS AND SCIENTIFIC INTEGRITY IN AI AND ADVANCED
TECHNOLOGIES

Deliverable:

Paradigm Justification Statement

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Research Topic

Explainable Artificial Intelligence Model for Early Prediction of the Risk of School Interruption among Secondary Education Students in Peru

1. Research Topic, Scope, and Context

School interruption remains an important challenge for the Peruvian educational system because students' continuity in secondary education can be affected by academic, socioeconomic, demographic, family, institutional, and territorial factors.

The Peruvian Ministry of Education has already developed initiatives such as *Alerta Escuela* to support the identification of students at risk of interrupting their studies. However, the proposed research does not intend merely to reproduce an existing risk-prediction mechanism. Instead, it seeks to investigate how an Explainable Artificial Intelligence (XAI) artifact can combine predictive capability with understandable explanations, stability, fairness, and responsible human use in the heterogeneous Peruvian educational context.

The proposed artifact is intended to support teachers and educational decision-makers rather than make autonomous decisions about students. A predicted risk score should therefore be interpreted as decision-support information and not as a deterministic label attached to a student.

The research will also consider the particular characteristics of the Peruvian educational context, including territorial heterogeneity and, where supported by the available data, differences related to region, rural or urban location, sex, home language, and other relevant demographic or socioeconomic variables.

2. Preliminary Research Question

How can an Explainable Artificial Intelligence artifact be designed and rigorously evaluated to support the early identification of the risk of school interruption among secondary education students in Peru while maintaining adequate predictive performance, faithful and stable explanations, and justifiable fairness criteria across relevant territorial and demographic groups?

This question deliberately emphasizes both **design** and **evaluation**. The intended contribution is therefore not only to identify statistical relationships associated with school interruption, but also to create and rigorously evaluate an AI artifact intended for responsible use in a real educational context.

3. Philosophical Position and Research Approach

The philosophical position adopted for this research is Pragmatism, while **Design Science Research (DSR)** is adopted as the primary research approach.

Pragmatism is appropriate because the value of the knowledge produced by the research is connected to its capacity to contribute to the understanding and solution of a relevant real-world problem. The research is not limited to observing educational risk; it seeks to design, evaluate, and refine an artifact intended to support human decision-making.

Design Science Research is appropriate because the principal doctoral contribution is expected to emerge from the construction and rigorous evaluation of an XAI artifact. Hevner et al. (2004) establish artifact creation, problem relevance, rigorous evaluation, research contribution, rigor, design as a search process, and communication as central principles of Design Science Research.

The study will therefore follow a build–evaluate–refine logic. Quantitative predictive evaluation and, if feasible and ethically approved, focused evaluation with educational stakeholders may be incorporated within the evaluation of the artifact without replacing DSR as the governing research approach.

- **Ontology — What is real?**

The research assumes that the risk of school interruption has observable manifestations and is associated with measurable educational, demographic, socioeconomic, and territorial conditions. However, the practical meaning and consequences of a risk prediction are context-dependent.

Therefore, the artifact should not assume that a predicted probability represents a deterministic description of a student. Risk is treated as an estimated condition based on available evidence rather than as a permanent characteristic of the individual.

- **Epistemology — What counts as valid knowledge?**

Valid knowledge will emerge from rigorous and reproducible evidence regarding the behavior of the proposed artifact.

This evidence may include predictive performance, calibration, temporal generalization, stability, fairness across relevant groups, and evaluation of the faithfulness and stability of the explanations produced by XAI techniques.

An important distinction will be maintained between **generating an explanation** and **demonstrating explainability**. Techniques such as SHAP or LIME can generate explanations, but additional evaluation is required to determine whether those explanations are faithful, stable, and potentially useful to intended users.

- **Methodology — How will knowledge be produced?**

Design Science Research will provide the governing methodological logic.

The research will involve identification and specification of the educational problem, definition of requirements for the artifact, development of predictive models, incorporation of XAI mechanisms, evaluation against appropriate baselines, and iterative refinement.

The empirical evaluation is expected to include supervised Machine Learning techniques, temporal validation when data permit it, analysis of class imbalance, calibration, precision-recall analysis, reproducibility controls, explanation stability and faithfulness assessment, and fairness analysis across selected groups.

The research will avoid claiming causal effects from predictive evidence alone. Demonstrating that an artifact predicts risk does not by itself demonstrate that its implementation reduces school interruption.

- **Axiology — What values guide the research?**

The research will prioritize fairness, privacy, transparency, human agency, non-maleficence, and responsible use of educational data.

Because educational information concerning minors may be involved, the study will consider appropriate authorization for data access, de-identification, secure storage, data minimization, and compliance with applicable Peruvian personal-data requirements, including **Ley N.º 29733, Ley de Protección de Datos Personales**.

Particular attention will be given to the risk of stigmatization. A student identified as being at high risk must not be treated as destined to interrupt his or her studies. Predictions should support professional judgment and appropriate educational interventions rather than automated punitive or exclusionary decisions.

The intended operational principle is:

Prediction → Explanation → Human Interpretation → Appropriate Educational Intervention

and not:

Prediction → Automated Decision.

4. Why Design Science Research?

DSR is selected because the principal contribution of the research is expected to be an **evaluated technological artifact together with knowledge obtained from its design and evaluation**.

A purely predictive or quasi-experimental Machine Learning study could compare algorithms and determine their relative predictive performance, but it would not fully represent the intended artifact-centered contribution.

A purely interpretivist study could provide valuable knowledge about how teachers and other stakeholders understand educational risk and AI explanations, but it would not by itself establish the technical performance of the artifact.

A true experimental design would be particularly appropriate if the principal research question were whether the actual use of the system **causes a reduction in school interruption**. That is not the principal causal claim of the current research.

Consequently, DSR provides the most coherent governing approach because the research seeks to **design, build, evaluate, and refine** an XAI artifact addressing a relevant educational problem.

5. Evaluation and Research Rigor

Predictive performance alone will not be considered sufficient evidence of artifact quality.

The evaluation should distinguish among several dimensions:

Predictive performance: ability to discriminate students with different observed outcomes.

Calibration: correspondence between predicted probabilities and observed outcomes.

Temporal validity: where feasible, training on earlier cohorts and evaluating on later cohorts to approximate prospective use.

Explanation faithfulness: whether explanations accurately represent the behavior of the predictive model.

Explanation stability: whether explanations remain reasonably consistent under controlled changes in seeds, samples, perturbations, or model configurations.

Fairness: evaluation of explicitly defined fairness criteria across relevant protected or contextually important attributes supported by the dataset.

Reproducibility: documentation of data-processing procedures, model configurations, random seeds, evaluation protocols, and relevant computational conditions.

The study will also distinguish the number of student observations from the number of independent model-training or evaluation runs. A large dataset does not eliminate the need to evaluate variability across model executions when claims of model superiority are made.

6. Relationship with Alerta Escuela and Intended Contribution

Alerta Escuela represents a natural reference point for positioning the proposed research in the Peruvian context.

The thesis should therefore explicitly determine whether the proposed artifact will **complement, benchmark against, or improve particular capabilities associated with existing early-warning approaches**, depending on the availability of comparable information and methodological feasibility.

The intended contribution is not simply another predictive model. The research seeks to investigate the combination of:

predictive capability + explainability + explanation stability + fairness + contextual adequacy + responsible human use.

The exact Design Science contribution category—such as **improvement** or **exaptation** according to Gregor and Hevner (2013)—will be established after the systematic review determines more precisely what is already known and what constitutes the defensible research gap.

7. Remaining Research Tension

The principal current methodological tension concerns the **granularity, temporal coverage, quality, and representativeness of the available educational data**.

Some accessible datasets may contain aggregated information rather than longitudinal student-level records. If adequate individual-level and temporally structured data cannot be obtained, the intended prediction target, fairness attributes, prediction horizon, or scope of the artifact may need to be revised.

A second related challenge is the precise operational definition of **school interruption**. The research must establish what event constitutes interruption and how far in advance the artifact is expected to predict it before model development and evaluation are finalized.

These limitations will be treated as design constraints rather than hidden after the artifact has been developed.

References

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